

"SAMPLE_TYPE"	"COLUMN_NAME"	"LABEL"	"LEGAL_VALUES"
"DESIGN"	"AG_EC09"	"NARS 9-level reporting region, based on aggregated Omernik Level III ecoregions. No missing or \"blank\" values allowed."	"CPL NAP NPL SAP SPL TPL UMW WMT XER "
"LAND"	"AG2006HISLP_WS_PCTFULL"	"Percent of area with data for Percentage of Agricultural Land (NLCD 2006) that lies on slopes greater than 20 degrees within the watershed (Ws)."	
"LAND"	"AG2006MIDSLP_WS_PCTFULL"	"Percent of area with data for Percentage of Agricultural Land (NLCD 2006) that lies on slopes greater than 10 degrees within the watershed (Ws)."	
"LAND"	"AGKFFACT_WS_PCTFULL"	"Percent of area with data for Mean surface soil erodibility factors adjusted (KFFACT) on agricultural land (NLCD 2006), within the upstream watershed (Ws)."	
"LAND"	"AGKFFACTWS"	"Mean surface soil erodibility factors adjusted (KFFACT) on agricultural land (NLCD 2006), within the upstream watershed (Ws)."	
"LAND"	"AL203_WS_PCTFULL"	"Percent of area with data for Mean of all Al203 values within the upstream watershed (Ws)."	
"LAND"	"AL203WS"	"Mean of all Al203 values within the upstream watershed (Ws)."	
"LAND"	"AWCH_WS_PCTFULL"	"Percent of area with data for Mean high values of available water capacity of soils (fraction) within the upstream watershed (Ws). From State Soil Geographic (STATSGO) Database."	
"LAND"	"AWCHWS"	"Mean high values of available water capacity of soils (fraction) within the upstream watershed (Ws). From State Soil Geographic (STATSGO) Database."	
"LAND"	"BDH_WS_PCTFULL"	"Percent of area with data for Mean high values of soil bulk density of soil types within the upstream watershed (Ws) (grams per cubic centimeter) from State Soil Geographic (STATSGO) Database."	
"LAND"	"BDHWS"	"Mean high values of soil bulk density of soil types within the upstream watershed (Ws) (grams per cubic centimeter) from State Soil Geographic (STATSGO) Database."	
"LAND"	"BEDROCK_PERM_WS_PCTFULL"	"Percent of area with data for bedrock permeability, based on principal aquifer permeability classes. "	
"LAND"	"BFI_WS_PCTFULL"	"Percent of area with data for Bfi is the ratio of base flow to total flow, expressed as a percentage. %, within the upstream watershed (Ws)."	
"LAND"	"BFIWS"	"Bfi is the ratio of base flow to total flow, expressed as a percentage. %, within the upstream watershed (Ws)."	
"LAND"	"CANALDENSITY_WS_PCTFULL"	"Percent of area with data for density of NHDPlus line features classified as canal, ditch, or pipeline within the upstream watershed (Ws). "	
"LAND"	"CANALDENSWS"	"Density of NHDPlus line features classified as canal, ditch, or pipeline within the upstream watershed (Ws). "	
"LAND"	"CAO_WS_PCTFULL"	"Percent of area with data for Mean of all CaO values within the upstream watershed (Ws)."	
"LAND"	"CAOWS"	"Mean of all CaO values within the upstream watershed (Ws)."	
"LANDSCAPE"	"CAT_COMID"	"Catchment COMID that the site falls in, from NHDPlusV21."	
"LAND"	"CATAREASQKM"	"Area in square kilometers of the sub-catchment created by accumulating from the survey site upstream within the NHDPlusV21 catchment."	
"LAND"	"CATAREASQKMRP100"	"Area in square kilometers within 100-m buffer of NHD stream lines (Rp100) of the sub-catchment created by accumulating from the survey site upstream within the NHDPlusV21 catchment."	
"LAND"	"CBNF_WS_PCTFULL"	"Percent of area with data for Mean rate of biological nitrogen fixation from the cultivation of crops in kg N/ha/yr, within the upstream watershed (Ws)."	
"LAND"	"CBNFWs"	"Mean rate of biological nitrogen fixation from the cultivation of crops in kg N/ha/yr, within the upstream watershed (Ws)."	
"LAND"	"CLAY_WS_PCTFULL"	"Percent of area with data for Mean of all clay values within the upstream watershed (Ws)."	
"LAND"	"CLAYWS"	"Mean of all clay values within the upstream watershed (Ws)."	
"LAND"	"COALMINEDENSWS"	"Density of georeferenced coal mine sites (USTRAT.shp) within the upstream watershed (Ws)."	
"LAND"	"COALMINES_WS_PCTFULL"	"Percent of area with data for density of georeferenced coal mine sites (USTRAT.shp) within the upstream watershed (Ws)."	
"LAND"	"COMPSTRGTH_WS_PCTFULL"	"Percent of area with data for Mean of all CompStrgth values within the upstream watershed (Ws)."	
"LAND"	"COMPSTRGTHWS"	"Mean of all CompStrgth values within the upstream watershed (Ws)."	
"LAND"	"DAMDENSWS"	"Density of georeferenced dams within the upstream watershed (Ws)."	
"LAND"	"DAMNIDSTORWS"	"Volume all reservoirs (NID_STORA in NID) per unit area of the upstream watershed (Ws). "	
"LAND"	"DAMNRMSTORWS"	"Volume all reservoirs (NORM_STORA in NID) per unit area of the upstream watershed (Ws)."	
"LAND"	"DAMS_WS_PCTFULL"	"Percent of area with data for Density of georeferenced dams within the upstream watershed (Ws)."	

"LAND" "DRAINAGEID" "Hydreoregion of the Drainage Basin that the site falls in, from NHDPlusV21."
"LANDSCAPE" "ELEV_PT" "Elevation at site location (meters) from 30m NHDPlusV2 which uses National Elevation Data (NED) snapshots from 2010 and 2011"
"LANDSCAPE" "ELEV_WS_MAX" "Maximum of all elev_cm values within the upstream watershed (Ws) divided by 100 to convert cm to m."
"LANDSCAPE" "ELEV_WS_MIN" "Minimum of all elev_cm values within the upstream watershed (Ws) divided by 100 to convert cm to m."
"LAND" "ELEV_WS_PCTFULL" "Percent of area with data for Mean of all elev_cm values within the upstream watershed (Ws) divided by 100 to convert cm to m."
"LAND" "ELEVWS" "Mean of all elev_cm values within the upstream watershed (Ws) divided by 100 to convert cm to m."
"LAND" "ET_WS_PCTFULL" "Percent of area with data for Mean-annual actual evapotranspiration (ET), estimated using regression equation of Sanford and Selnick (2013)"
"LAND" "ETWS" "Mean-annual actual evapotranspiration (ET), estimated using regression equation of Sanford and Selnick (2013)"
"LAND" "FE203_WS_PCTFULL" "Percent of area with data for Mean of all Fe203 values within the upstream watershed (Ws)."
"LAND" "FE203WS" "Mean of all Fe203 values within the upstream watershed (Ws)."
"LAND" "FERT_WS_PCTFULL" "Percent of area with data for Mean rate of synthetic nitrogen fertilizer application to agricultural land in kg N/ha/yr, within the upstream watershed (Ws)."
"LAND" "FERTWS" "Mean rate of synthetic nitrogen fertilizer application to agricultural land in kg N/ha/yr, within the upstream watershed (Ws)."
"LAND" "FIRE2000_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2000 within the upstream watershed (Ws)."
"LAND" "FIRE2000_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2000 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2001_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2001 within the upstream watershed (Ws)."
"LAND" "FIRE2001_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2001 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2002_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2002 within the upstream watershed (Ws)."
"LAND" "FIRE2002_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2002 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2003_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2003 within the upstream watershed (Ws)."
"LAND" "FIRE2003_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2003 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2004_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2004 within the upstream watershed (Ws)."
"LAND" "FIRE2004_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2004 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2005_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2005 within the upstream watershed (Ws)."
"LAND" "FIRE2005_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2005 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2006_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2006 within the upstream watershed (Ws)."
"LAND" "FIRE2006_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2006 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2007_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2007 within the upstream watershed (Ws)."
"LAND" "FIRE2007_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2007 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."
"LAND" "FIRE2008_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2008 within the upstream watershed (Ws)."
"LAND" "FIRE2008_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire

perimeter) for year 2008 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "FIRE2009_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2009 within the upstream watershed (Ws)."

"LAND" "FIRE2009_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2009 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "FIRE2010_WS_PCTFULL" "Percent of area with data for % Forest loss to fire (fire perimeter) for 2010 within the upstream watershed (Ws)."

"LAND" "FIRE2010_WS_PCTFULLRP100" "Percent of area with data for % Forest loss to fire (fire perimeter) for year 2010 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "FSTFRZ_PT" "Site mean day of the year of first freeze, derived from 30 years of record (1961-1990), 2km PRISM. For example, value of 300 is the 300th day of the year (Oct 27th). Values > 365 mean that the first freeze occurs after Dec 31. It is done this way so that (first_freeze - last_freeze) should always give a positive number; high numbers for an area with a long frost-free period (e.g. Florida), low numbers for an area with a short frost-free period."

"LAND" "FSTFRZ_WS_PCTFULL" "Percent of area with data for Site average of mean day of the year of first freeze, derived from 30 years of record (1961-1990), 2km PRISM. For example, value of 300 is the 300th day of the year (Oct 27th). Values > 365 mean that the first freeze occurs after Dec 31. It is done this way so that (first_freeze - last_freeze) should always give a positive number; high numbers for an area with a long frost-free period (e.g. Florida), low numbers for an area with a short frost-free period."

"LAND" "FSTFRZWS" "Site average of mean day of the year of first freeze, derived from 30 years of record (1961-1990), 2km PRISM. For example, value of 300 is the 300th day of the year (Oct 27th). Values > 365 mean that the first freeze occurs after Dec 31. It is done this way so that (first_freeze - last_freeze) should always give a positive number; high numbers for an area with a long frost-free period (e.g. Florida), low numbers for an area with a short frost-free period."

"LAND" "GEOL_HUNT_DOM_PCT_WS" "Dominant (highest percent of area) geology, derived from Hunt (1979) coverage"

"LANDSCAPE" "GEOL_HUNT_DOM_WS" "Percentage of the watershed covered by the dominant geology type (Hunt, 1979)"

"LANDSCAPE" "GEOL_HUNT_PT" "Value of categorized geology at site location, derived from a simplified version of Hunt (1979)."

"LAND" "GEOL_HUNT_WS_PCTFULL" "Percent of area with data for categorized geology (Hunt, 1979) in watershed."

"LAND" "GEOL_REEDBUSH_DOM_PCT_WS" "Percentage of the watershed covered by the dominant geology type (Reed & Bush, 2001)"

"LANDSCAPE" "GEOL_REEDBUSH_DOM_WS" "Dominant (highest percent of area) geology, derived from a simplified version of Reed & Bush (2001) - Generalized Geologic Map of the Conterminous United States."

"LANDSCAPE" "GEOL_REEDBUSH_PT" "Value of categorized geology at site location, derived from a simplified version of Reed & Bush (2001)."

"LAND" "GEOL_REEDBUSH_WS_PCTFULL" "Percent of area with data for categorized geology (Reed & Bush, 2001) in watershed."

"LAND" "GNIS_NAME" "Name of the stream reach that the site was snapped to from the NHDPlusV21."

"DESIGN" "GNIS_NAME" "USGS Geographic Names Information System name (from NHD). Value of \"None\" if no GNIS_NAME available. . No missing or \"blank\" values allowed."

"LAND" "HUDEN2010_WS_PCTFULL" "Percent of area with data for Mean of all housing density values (housing units / km2) within the upstream watershed."

"LAND" "HUDEN2010_WS_PCTFULLRP100" "Percent of area with data for Mean of all housing density values (housing units / km2) within the upstream watershed and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "HUDEN2010WS" "Mean of all housing density values (housing units / km2) within the upstream watershed."

"LAND" "HUDEN2010WSRP100" "Mean of all housing density values (housing units / km2) within the upstream watershed and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "HYDRLCOND_WS_PCTFULL" "Percent of area with data for Mean of all hydraulic conductivity (in micrometers per second) of surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "HYDRLCONDWS" "Mean of all hydraulic conductivity (in micrometers per second) of surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "INORGNWETDEP_2008_WS_PCTFULL" "Percent of area with data for Annual average of precipitation-weighted mean wet deposition for inorganic nitrogen concentration from nitrate and ammonium for year, 2008, within the upstream watershed (Ws)."

"LAND" "INORGNWETDEP_2008WS" "Annual average of precipitation-weighted mean wet deposition for inorganic nitrogen concentration from nitrate and ammonium for year, 2008, within the upstream watershed (Ws)."

"LAND" "K2O_WS_PCTFULL" "Percent of area with data for Mean of all lithological potassium oxide (K2O) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "K2OWS" "Mean of all lithological potassium oxide (K2O) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "KFFACT_PT" "Site mean of the soil erodibility factor (no units) of soils from the State Soil Geographic (STATSGO) Database."

"LAND" "KFFACT_WS_PCTFULL" "Percent of area with data for Mean surface soil erodibility factors adjusted (KFFACT) , within the upstream watershed (Ws)."

"LAND" "KFFACTWS" "Mean surface soil erodibility factors adjusted (KFFACT) , within the upstream watershed (Ws)."

"LAND" "LITH_WS_PCTFULL" "Percent of area with data for lithology type metrics for the watershed."

"LANDSCAPE" "LSTFRZ_PT" "Site average of mean day of the year of last freeze, derived from 30 years of record (1961-1990), 2km PRISM. For example, value of 100 is the 100th day of the year (April 10th). Values of 0 here indicate there is no freeze"

"LAND" "LSTFRZ_WS_PCTFULL" "Percent of area with data for Watershed average of mean day of the year of last freeze, derived from 30 years of record (1961-1990), 2km PRISM. For example, value of 100 is the 100th day of the year (April 10th). Values of 0 here indicate there is no freeze"

"LAND" "LSTFRZWS" "Watershed average of mean day of the year of last freeze, derived from 30 years of record (1961-1990), 2km PRISM. For example, value of 100 is the 100th day of the year (April 10th). Values of 0 here indicate there is no freeze"

"LAND" "MANURE_WS_PCTFULL" "Percent of area with data for Mean rate of manure application to agricultural land from confined animal feeding operations in kg N/ha/yr, within the upstream watershed (Ws)."

"LAND" "MANUREWS" "Mean rate of manure application to agricultural land from confined animal feeding operations in kg N/ha/yr, within the upstream watershed (Ws)."

"LAND" "MAST_2018" "Predicted reference-condition mean annual stream temperatures. These temperatures are predictions of what stream temperatures could be under minimal human-related alterations to watersheds - not current conditions"

"LAND" "MAST_2019" "Predicted reference-condition mean annual stream temperatures. These temperatures are predictions of what stream temperatures could be under minimal human-related alterations to watersheds - not current conditions"

"LAND" "MGO_WS_PCTFULL" "Percent of area with data for Mean of all lithological magnesium oxide (MgO) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "MGOWS" "Mean of all lithological magnesium oxide (MgO) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "MINE_WS_PCTFULL" "Percent of area with data for density of georeferenced mine sites within the upstream watershed (Ws)."

"LAND" "MINE_WS_PCTFULLRP100" "Percent of area with data for density of georeferenced mine sites within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "MINEDENSWS" "Density of georeferenced mine sites within the upstream watershed (Ws)."

"LAND" "MINEDENSWSRP100" "Density of georeferenced mine sites within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "MSST_2018" "Predicted reference-condition mean summer stream temperatures. These temperatures are predictions of what stream temperatures could be under minimal human-related alterations to watersheds - not current conditions"

"LAND" "MSST_2019" "Predicted reference-condition mean summer stream temperatures. These temperatures are predictions of what stream temperatures could be under minimal human-related alterations to watersheds - not current conditions"

"LAND" "MWST_2018" "Predicted reference-condition mean winter stream temperatures. These temperatures are predictions of what stream temperatures could be under minimal human-related alterations to watersheds - not current conditions"

"LAND" "MWST_2019" "Predicted reference-condition mean winter stream temperatures. These temperatures are predictions of what stream temperatures could be under minimal human-related alterations to watersheds - not current conditions"

"LAND" "N_WS_PCTFULL" "Percent of area with data for Mean of all lithological nitrogen (N) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "NA2O_WS_PCTFULL" "Percent of area with data for Mean of all lithological sodium oxide (Na2O) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "NA2OWS" "Mean of all lithological sodium oxide (Na2O) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "NABD_DENSWS" "Density of georeferenced dams within the upstream watershed (Ws)."

"LAND" "NABD_NIDSTORWS" "Volume of all reservoirs (NID_stor in NABD) per unit area of the upstream watershed (Ws)."

"LAND" "NABD_NRMSTORWS" "Volume of all reservoirs (Norm_stor in NABD) per unit area of the upstream watershed (Ws)."

"LAND" "NABD_WS_PCTFULL" "Percent of area with data for density of georeferenced dams within the upstream watershed (Ws)."

"LAND" "NH4_2008_WS_PCTFULL" "Percent of area with data for Annual average of precipitation-weighted mean deposition for ammonium ion concentration for year, 2008, within the upstream watershed (Ws)."

"LAND" "NH4_2008WS" "Annual average of precipitation-weighted mean deposition for ammonium ion concentration for year, 2008, within the upstream watershed (Ws)."

"LAND" "NLCD2001_WS_PCTFULL" "Percent of area with data for NLCD 2001 for upstream watershed."

"LAND" "NLCD2001_WS_PCTFULLRP100" "Percent of area with data for NLCD 2001 for upstream watershed withing a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2004_WS_PCTFULL" "Percent of area with data for NLCD 2004 for upstream watershed."

"LAND" "NLCD2004_WS_PCTFULLRP100" "Percent of area with data for NLCD 2004 for upstream watershed within a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2006_WS_PCTFULL" "Percent of area with data for NLCD 2006 for upstream watershed."

"LAND" "NLCD2006_WS_PCTFULLRP100" "Percent of area with data for NLCD 2006 for upstream watershed withing a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2008_WS_PCTFULL" "Percent of area with data for NLCD 2008 for upstream watershed."

"LAND" "NLCD2008_WS_PCTFULLRP100" "Percent of area with data for NLCD 2008 for upstream watershed within a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2011_WS_PCTFULL" "Percent of area with data for NLCD 2011 for upstream watershed."

"LAND" "NLCD2011_WS_PCTFULLRP100" "Percent of area with data for NLCD 2011 for upstream watershed withing a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2013_WS_PCTFULL" "Percent of area with data for NLCD 2013 for upstream watershed."

"LAND" "NLCD2013_WS_PCTFULLRP100" "Percent of area with data for NLCD 2013 for upstream watershed within a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2016_WS_PCTFULL" "Percent of area with data for NLCD 2016 for upstream watershed."

"LAND" "NLCD2016_WS_PCTFULLRP100" "Percent of area with data for NLCD 2016 for upstream watershed withing a 100-m buffer of the NHD stream lines."

"LAND" "NLCD2019_WS_PCTFULL" "Percent of area with data for NLCD 2019 for upstream watershed."

"LAND" "NLCD2019_WS_PCTFULLRP100" "Percent of area with data for NLCD 2019 for upstream watershed within a 100-m buffer of the NHD stream lines."

"LAND" "NO3_2008_WS_PCTFULL" "Percent of area with data for Annual average of precipitation-weighted mean deposition for nitrate ion concentration for year, 2008, within the upstream watershed (Ws)."

"LAND" "NO3_2008WS" "Annual average of precipitation-weighted mean deposition for nitrate ion concentration for year, 2008, within the upstream watershed (Ws)."

"LAND" "NPDES_WS_PCTFULL" "Percent of area with data for density of georeferenced NPDES sites (NPDES_Major.shp) within the upstream watershed (Ws)."

"LAND" "NPDES_WS_PCTFULLRP100" "Percent of area with data for density of georeferenced NPDES sites (NPDES_Major.shp) within a 100-m buffer (Rp100) of the NHD stream lines. Derived as the intersection between RpRp100 and NPDES_Major.shp landscape layers."

"LAND" "NPDESDENSWS" "Density of georeferenced NPDES sites (NPDES_Major.shp) within the upstream watershed (Ws)."

"LAND" "NPDESDENSWSRP100" "Density of georeferenced NPDES sites (NPDES_Major.shp) within a 100-m buffer (Rp100) of the NHD stream lines. Derived as the intersection between RpRp100 and NPDES_Major.shp landscape layers."

"LAND" "NWS" "Mean of all lithological nitrogen (N) content in surface or near surface geology values within the upstream watershed (Ws)."

"LAND" "OM_WS_PCTFULL" "Percent of area with data for Mean of all om values within the upstream watershed (Ws)."

"LAND" "OMH_WS_PCTFULL" "Percent of area with data for Watershed mean of the high value of organic matter content (percent by weight) of soils from State Soil Geographic (STATSGO) Database."

"LAND" "OMHWS" "Watershed mean of the high value of organic matter content (percent by weight) of soils from State Soil Geographic (STATSGO) Database."

"LAND" "OMWS" "Mean of all om values within the upstream watershed (Ws)."

"LAND" "P205_WS_PCTFULL" "Percent of area with data for Mean of all P205 values within the upstream watershed (Ws)."

"LAND" "P205WS" "Mean of all P205 values within the upstream watershed (Ws)."

"LAND" "PCTAG2006SLP10WS" "Percentage of Agricultural Land (NLCD 2006)that lies on slopes greater than 10 degrees within the watershed (Ws)."

"LAND" "PCTAG2006SLP20WS" "Percentage of Agricultural Land (NLCD 2006) that lies on slopes

greater than 20 degrees within the watershed (Ws)."

"LAND" "PCTALKINTRUVOLWS" "% of the upstream watershed (Ws) classified as lithology type: alkaline intrusive volcanic rock."

"LAND" "PCTALLUVCOASTWS" "% of the upstream watershed (Ws) classified as lithology type: alluvium and fine-textured coastal zone sediment."

"LAND" "PCTBL2001_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 31) landscape raster."

"LAND" "PCTBL2001WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2001.tif (class 31) landscape raster."

"LAND" "PCTBL2004_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 31) landscape raster."

"LAND" "PCTBL2004WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2004.tif (class 31) landscape raster."

"LAND" "PCTBL2006_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 31) landscape raster."

"LAND" "PCTBL2006WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2006.tif (class 31) landscape raster."

"LAND" "PCTBL2008_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 31) landscape raster."

"LAND" "PCTBL2008WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2008.tif (class 31) landscape raster."

"LAND" "PCTBL2011_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 31) landscape raster."

"LAND" "PCTBL2011WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2011.tif (class 31) landscape raster."

"LAND" "PCTBL2013_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 31) landscape raster."

"LAND" "PCTBL2013WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2013.tif (class 31) landscape raster."

"LAND" "PCTBL2016_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 31) landscape raster."

"LAND" "PCTBL2016WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2016.tif (class 31) landscape raster."

"LAND" "PCTBL2019_WSRP100" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 31) landscape raster."

"LAND" "PCTBL2019WS" "% of the upstream watershed (Ws) areas classified as areas of bedrock, desert pavement, scarps, talus, slides, volcanic material, glacial debris, sand dunes, strip mines, gravel pits, and other accumulations of earthen material. Derived from the nlcd2019.tif (class 31) landscape raster."

"LAND" "PCTCARBONATEWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Carbonate rock."

"LAND" "PCTCARBRESIDWS" "% of the upstream watershed (Ws) classified as lithology type: carbonate residual material."

"LAND" "PCTCOASTCRSWS" "% of the upstream watershed (Ws) classified as lithology type: coastal zone sediment, coarse-textured."

"LAND" "PCTCOLLUVSEDWS" "% of the upstream watershed (Ws) classified as lithology type: colluvial sediment."

"LAND" "PCTCONIF2001_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 42) landscape raster."

"LAND" "PCTCONIF2001WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2001.tif (class 42) landscape raster."

"LAND" "PCTCONIF2004_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 42) landscape raster."

"LAND" "PCTCONIF2004WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2004.tif (class 42) landscape raster."

"LAND" "PCTCONIF2006_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 42) landscape raster."

"LAND" "PCTCONIF2006WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2006.tif (class 42) landscape raster."

"LAND" "PCTCONIF2008_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 42) landscape raster."

"LAND" "PCTCONIF2008WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2008.tif (class 42) landscape raster."

"LAND" "PCTCONIF2011_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 42) landscape raster."

"LAND" "PCTCONIF2011WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2011.tif (class 42) landscape raster."

"LAND" "PCTCONIF2013_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 42) landscape raster."

"LAND" "PCTCONIF2013WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2013.tif (class 42) landscape raster."

"LAND" "PCTCONIF2016_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 42) landscape raster."

"LAND" "PCTCONIF2016WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2016.tif (class 42) landscape raster."

"LAND" "PCTCONIF2019_WSRP100" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 42) landscape raster."

"LAND" "PCTCONIF2019WS" "% of the upstream watershed (Ws) areas classified as evergreen forest land cover. Derived from the nlcd2019.tif (class 42) landscape raster."

"LAND" "PCTCROP2001_WSRP100" "% of the upstream watershed (Ws) areas classified as row crop land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 82) landscape raster"

"LAND" "PCTCROP2001WS" "% of the upstream watershed (Ws) areas classified as row crop land use. Derived from the nlcd2001.tif (class 82) landscape raster"

"LAND" "PCTCROP2004_WSRP100" "% of the upstream watershed (Ws) areas classified as row crop land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 82) landscape raster"

"LAND" "PCTCROP2004WS" "% of the upstream watershed (Ws) areas classified as row crop land use. Derived from the nlcd2004.tif (class 82) landscape raster"

"LAND" "PCTCROP2006_WSRP100" "% of the upstream watershed (Ws) areas classified as row crop land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 82) landscape raster"

[illegible]

"LAND" "PCTDECID2019WS" "% of the upstream watershed (Ws) areas classified as deciduous forest land cover. Derived from the nlcd2019.tif (class 41) landscape raster."

"LAND" "PCTEOLCRSWS" "% of the upstream watershed (Ws) classified as lithology type: eolian sediment, coarse-textured (sand dunes)."

"LAND" "PCTEOLFINEWS" "% of the upstream watershed (Ws) classified as lithology type: eolian sediment, fine-textured (glacial loess)."

"LAND" "PCTEXTRUVOLWS" "% of the upstream watershed (Ws) classified as lithology type: extrusive volcanic rock."

"LAND" "PCTFIRE2000WS" " % Forest loss to fire (fire perimeter) for 2000 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2000WSRP100" " % Forest loss to fire (fire perimeter) for year 2000 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2001WS" "% Forest loss to fire (fire perimeter) for 2001 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2001WSRP100" "% Forest loss to fire (fire perimeter) for year 2001 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2002WS" "% Forest loss to fire (fire perimeter) for 2002 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2002WSRP100" "% Forest loss to fire (fire perimeter) for year 2002 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2003WS" "% Forest loss to fire (fire perimeter) for 2003 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2003WSRP100" "% Forest loss to fire (fire perimeter) for year 2003 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2004WS" "% Forest loss to fire (fire perimeter) for 2004 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2004WSRP100" "% Forest loss to fire (fire perimeter) for year 2004 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2005WS" "% Forest loss to fire (fire perimeter) for 2005 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2005WSRP100" "% Forest loss to fire (fire perimeter) for year 2005 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2006WS" "% Forest loss to fire (fire perimeter) for 2006 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2006WSRP100" "% Forest loss to fire (fire perimeter) for year 2006 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2007WS" "% Forest loss to fire (fire perimeter) for 2007 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2007WSRP100" "% Forest loss to fire (fire perimeter) for year 2007 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2008WS" "% Forest loss to fire (fire perimeter) for 2008 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2008WSRP100" "% Forest loss to fire (fire perimeter) for year 2008 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2009WS" "% Forest loss to fire (fire perimeter) for 2009 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2009WSRP100" "% Forest loss to fire (fire perimeter) for year 2009 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFIRE2010WS" "% Forest loss to fire (fire perimeter) for 2010 within the upstream watershed (Ws)."

"LAND" "PCTFIRE2010WSRP100" "% Forest loss to fire (fire perimeter) for year 2010 within the upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100)."

"LAND" "PCTFRSTLOSS2001WS" "% Forest cover loss (Tree canopy cover change) for 2001 within watershed"

"LAND" "PCTFRSTLOSS2001WSRP100" "% Forest cover loss (Tree canopy cover change) for 2001 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2002WS" "% Forest cover loss (Tree canopy cover change) for 2002 within watershed"

"LAND" "PCTFRSTLOSS2002WSRP100" "% Forest cover loss (Tree canopy cover change) for 2002 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2003WS" "% Forest cover loss (Tree canopy cover change) for 2003 within watershed"

"LAND" "PCTFRSTLOSS2003WSRP100" "% Forest cover loss (Tree canopy cover change) for 2003 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2004WS" "% Forest cover loss (Tree canopy cover change) for 2004 within watershed"

watershed"

"LAND" "PCTFRSTLOSS2004WSRP100" "% Forest cover loss (Tree canopy cover change) for 2004 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2005WS" "% Forest cover loss (Tree canopy cover change) for 2005 within watershed"

"LAND" "PCTFRSTLOSS2005WSRP100" "% Forest cover loss (Tree canopy cover change) for 2005 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2006WS" "% Forest cover loss (Tree canopy cover change) for 2006 within watershed"

"LAND" "PCTFRSTLOSS2006WSRP100" "% Forest cover loss (Tree canopy cover change) for 2006 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2007WS" "% Forest cover loss (Tree canopy cover change) for 2007 within watershed"

"LAND" "PCTFRSTLOSS2007WSRP100" "% Forest cover loss (Tree canopy cover change) for 2007 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2008WS" "% Forest cover loss (Tree canopy cover change) for 2008 within watershed"

"LAND" "PCTFRSTLOSS2008WSRP100" "% Forest cover loss (Tree canopy cover change) for 2008 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2009WS" "% Forest cover loss (Tree canopy cover change) for 2009 within watershed"

"LAND" "PCTFRSTLOSS2009WSRP100" "% Forest cover loss (Tree canopy cover change) for 2009 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2010WS" "% Forest cover loss (Tree canopy cover change) for 2010 within watershed"

"LAND" "PCTFRSTLOSS2010WSRP100" "% Forest cover loss (Tree canopy cover change) for 2010 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2011WS" "% Forest cover loss (Tree canopy cover change) for 2011 within watershed"

"LAND" "PCTFRSTLOSS2011WSRP100" "% Forest cover loss (Tree canopy cover change) for 2011 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2012WS" "% Forest cover loss (Tree canopy cover change) for 2012 within watershed"

"LAND" "PCTFRSTLOSS2012WSRP100" "% Forest cover loss (Tree canopy cover change) for 2012 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSS2013WS" "% Forest cover loss (Tree canopy cover change) for 2013 within watershed"

"LAND" "PCTFRSTLOSS2013WSRP100" "% Forest cover loss (Tree canopy cover change) for 2013 within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTFRSTLOSSBYYEAR_WS_PCTFULL" "Percent of area with data for % Forest cover loss (Tree canopy cover change) within watershed"

"LAND" "PCTFRSTLOSSBYYEAR_WS_PCTFULLRP100" "Percent of area with data for % Forest cover loss (Tree canopy cover change) within watershed and within 100-m buffer of NHD stream lines"

"LAND" "PCTGLACLAKECRSWS" "% of the upstream watershed (Ws) classified as lithology type: glacial outwash and glacial lake sediment, coarse-textured."

"LAND" "PCTGLACLAKEFINEWS" "% of the upstream watershed (Ws) classified as lithology type: glacial outwash and glacial lake sediment, fine-textured."

"LAND" "PCTGLACTILCLAYWS" "% of the upstream watershed (Ws) classified as lithology type: glacial till, clayey."

"LAND" "PCTGLACTILCRSWS" "% of the upstream watershed (Ws) classified as lithology type: glacial till, coarse-textured."

"LAND" "PCTGLACTILLOAMWS" "% of the upstream watershed (Ws) classified as lithology type: glacial till, loamy."

"LAND" "PCTGRS2001_WSRP100" "% of the upstream watershed (Ws) areas classified as grassland/herbaceous land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 71) landscape raster."

"LAND" "PCTGRS2001WS" "% of the upstream watershed (Ws) areas classified as grassland/herbaceous land cover. Derived from the nlcd2001.tif (class 71) landscape raster."

"LAND" "PCTGRS2004_WSRP100" "% of the upstream watershed (Ws) areas classified as grassland/herbaceous land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 71) landscape raster."

"LAND" "PCTGRS2004WS" "% of the upstream watershed (Ws) areas classified as grassland/herbaceous land cover. Derived from the nlcd2004.tif (class 71) landscape raster."

"LAND" "PCTGRS2006_WSRP100" "% of the upstream watershed (Ws) areas classified as grassland/herbaceous land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the

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81) landscape raster."

"LAND" "PCTHAY2019WS" "% of the upstream watershed (Ws) areas classified as pasture/hay land use. Derived from the nlcd2019.tif (class 81) landscape raster."

"LAND" "PCTHBWET2001_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 95) landscape raster."

"LAND" "PCTHBWET2001WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2001.tif (class 95) landscape raster."

"LAND" "PCTHBWET2004_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 95) landscape raster."

"LAND" "PCTHBWET2004WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2004.tif (class 95) landscape raster."

"LAND" "PCTHBWET2006_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 95) landscape raster."

"LAND" "PCTHBWET2006WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2006.tif (class 95) landscape raster."

"LAND" "PCTHBWET2008_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 95) landscape raster."

"LAND" "PCTHBWET2008WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2008.tif (class 95) landscape raster."

"LAND" "PCTHBWET2011_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 95) landscape raster."

"LAND" "PCTHBWET2011WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2011.tif (class 95) landscape raster."

"LAND" "PCTHBWET2013_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 95) landscape raster."

"LAND" "PCTHBWET2013WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2013.tif (class 95) landscape raster."

"LAND" "PCTHBWET2016_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 95) landscape raster."

"LAND" "PCTHBWET2016WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2016.tif (class 95) landscape raster."

"LAND" "PCTHBWET2019_WSRP100" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 95) landscape raster."

"LAND" "PCTHBWET2019WS" "% of the upstream watershed (Ws) areas classified as herbaceous wetland land cover. Derived from the nlcd2019.tif (class 95) landscape raster."

"LAND" "PCTHYDRICWS" "% of the upstream watershed (Ws) classified as lithology type: hydric, peat and muck."

"LAND" "PCTICE2001_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 12) landscape raster."

"LAND" "PCTICE2001WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2001.tif (class 12) landscape raster."

"LAND" "PCTICE2004_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 12) landscape raster."

"LAND" "PCTICE2004WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2004.tif (class 12) landscape raster."

"LAND" "PCTICE2006_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 12) landscape raster."

"LAND" "PCTICE2006WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2006.tif (class 12) landscape raster."

"LAND" "PCTICE2008_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 12) landscape raster."

"LAND" "PCTICE2008WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2008.tif (class 12) landscape raster."

"LAND" "PCTICE2011_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 12) landscape raster."

"LAND" "PCTICE2011WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2011.tif (class 12) landscape raster."

"LAND" "PCTICE2013_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 12) landscape raster."

"LAND" "PCTICE2013WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2013.tif (class 12) landscape raster."

"LAND" "PCTICE2016_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 12) landscape raster."

"LAND" "PCTICE2016WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2016.tif (class 12) landscape raster."

"LAND" "PCTICE2019_WSRP100" "% of the upstream watershed (Ws) areas classified as ice/snow land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 12) landscape raster."

"LAND" "PCTICE2019WS" "% of the upstream watershed (Ws) areas classified as ice/snow land cover. Derived from the nlcd2019.tif (class 12) landscape raster."

"LAND" "PCTIMP2001_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2001) within the upstream watershed (Ws)."

"LAND" "PCTIMP2001_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2001) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2001WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2001) within the upstream watershed (Ws)."

"LAND" "PCTIMP2001WSRP100" "Mean imperviousness of anthropogenic surfaces (NLCD 2001) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2004_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2004) within the upstream watershed (Ws)."

"LAND" "PCTIMP2004_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2004) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2004WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2004) within the upstream watershed (Ws)."

"LAND" "PCTIMP2004WSRP100" "Mean imperviousness of anthropogenic surfaces (NLCD 2004) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2006_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2006) within the upstream watershed (Ws)."

"LAND" "PCTIMP2006_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2006) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2006WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2006) within the upstream watershed (Ws)."

"LAND" "PCTIMP2006WSRP100" "Mean imperviousness of anthropogenic surfaces (NLCD 2006) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2008_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2008) within the upstream watershed (Ws)."

"LAND" "PCTIMP2008_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2008) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2008WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2008) within the upstream watershed (Ws)."

"LAND" "PCTIMP2008WSRP100" "Mean imperviousness of anthropogenic surfaces (NLCD 2008) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2011_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2011) within the upstream watershed (Ws)."

"LAND" "PCTIMP2011_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2011) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2011WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2011) within the upstream watershed (Ws)."

"LAND" "PCTIMP2011WSRP100" "Mean imperviousness of anthropogenic surfaces (NLCD 2011) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2013_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2013) within the upstream watershed (Ws)."

"LAND" "PCTIMP2013_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2013) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2013_WSRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2013) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2013WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2013) within the upstream watershed (Ws)."

"LAND" "PCTIMP2016_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2016) within the upstream watershed (Ws)."

"LAND" "PCTIMP2016_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2016) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2016WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2016) within the upstream watershed (Ws)."

"LAND" "PCTIMP2016WSRP100" "Mean imperviousness of anthropogenic surfaces (NLCD 2016) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2019_WS_PCTFULL" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2019) within the upstream watershed (Ws)."

"LAND" "PCTIMP2019_WS_PCTFULLRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2019) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2019_WSRP100" "Percent of area with data for Mean imperviousness of anthropogenic surfaces (NLCD 2019) within watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTIMP2019WS" "Mean imperviousness of anthropogenic surfaces (NLCD 2019) within the upstream watershed (Ws)."

"LAND" "PCTMXFST2001_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 43) landscape raster."

"LAND" "PCTMXFST2001WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2001.tif (class 43) landscape raster."

"LAND" "PCTMXFST2004_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 43) landscape raster."

"LAND" "PCTMXFST2004WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2004.tif (class 43) landscape raster."

"LAND" "PCTMXFST2006_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 43) landscape raster."

"LAND" "PCTMXFST2006WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2006.tif (class 43) landscape raster."

"LAND" "PCTMXFST2008_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 43) landscape raster."

"LAND" "PCTMXFST2008WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2008.tif (class 43) landscape raster."

"LAND" "PCTMXFST2011_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 43) landscape raster."

"LAND" "PCTMXFST2011WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2011.tif (class 43) landscape raster."

"LAND" "PCTMXFST2013_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 43) landscape raster."

"LAND" "PCTMXFST2013WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2013.tif (class 43) landscape raster."

"LAND" "PCTMXFST2016_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 43) landscape raster."

"LAND" "PCTMXFST2016WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2016.tif (class 43) landscape raster."

"LAND" "PCTMXFST2019_WSRP100" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 43) landscape raster."

"LAND" "PCTMXFST2019WS" "% of the upstream watershed (Ws) areas classified as mixed deciduous/evergreen forest land cover. Derived from the nlcd2019.tif (class 43) landscape raster."

"LAND" "PCTNONAGINTRODMANAGVEG_WS_PCTFULL" "Percent of area with data for % Nonnative vegetation landcover type reclassified from LANDFIRE Existing Vegetation Type (EVT), within upstream watershed (Ws). "

"LAND" "PCTNONAGINTRODMANAGVEG_WS_PCTFULLRP100" "Percent of area with data for % Nonnative vegetation landcover type reclassified from LANDFIRE Existing Vegetation Type (EVT), within upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTNONAGINTRODMANAGVEGWS" "% Nonnative vegetation landcover type reclassified from LANDFIRE Existing Vegetation Type (EVT), within upstream watershed (Ws). "

"LAND" "PCTNONAGINTRODMANAGVEGWSRP100" "% Nonnative vegetation landcover type reclassified from LANDFIRE Existing Vegetation Type (EVT), within upstream watershed (Ws) and within 100-m buffer of NHD stream lines (Rp100). "

"LAND" "PCTNONCARBRESIDWS" "% of the upstream watershed (Ws) classified as lithology type: non-carbonate residual material."

"LAND" "PCTNOT_PRINCIPLEWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Not a principal aquifer."

"LAND" "PCTOW2001_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 11) landscape raster."

"LAND" "PCTOW2001WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2001.tif (class 11) landscape raster."

"LAND" "PCTOW2004_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 11) landscape raster."

"LAND" "PCTOW2004WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2004.tif (class 11) landscape raster."

"LAND" "PCTOW2006_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 11) landscape raster."

"LAND" "PCTOW2006WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2006.tif (class 11) landscape raster."

"LAND" "PCTOW2008_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 11) landscape raster."

"LAND" "PCTOW2008WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2008.tif (class 11) landscape raster."

"LAND" "PCTOW2011_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 11) landscape raster."

"LAND" "PCTOW2011WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2011.tif (class 11) landscape raster."

"LAND" "PCTOW2013_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 11) landscape raster."

"LAND" "PCTOW2013WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2013.tif (class 11) landscape raster."

"LAND" "PCTOW2016_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 11) landscape raster."

"LAND" "PCTOW2016WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2016.tif (class 11) landscape raster."

"LAND" "PCTOW2019_WSRP100" "% of the upstream watershed (Ws) areas classified as open water land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 11) landscape raster."

"LAND" "PCTOW2019WS" "% of the upstream watershed (Ws) areas classified as open water land cover. Derived from the nlcd2019.tif (class 11) landscape raster."

"LAND" "PCTSALLAKEWS" "% of the upstream watershed (Ws) classified as lithology type: saline like sediment."

"LAND" "PCTSANDSTONE_CARBONATEWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Sandstone and carbonate rocks."

"LAND" "PCTSANDSTONEWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Sandstone."

"LAND" "PCTSEMICONCONSOL_SANDWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Semiconsolidated sand."

"LAND" "PCTSHRB2001_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 52) landscape raster."

"LAND" "PCTSHRB2001WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2001.tif (class 52) landscape raster."

"LAND" "PCTSHRB2004_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 52) landscape raster."

"LAND" "PCTSHRB2004WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2004.tif (class 52) landscape raster."

"LAND" "PCTSHRB2006_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 52) landscape raster."

"LAND" "PCTSHRB2006WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2006.tif (class 52) landscape raster."

"LAND" "PCTSHRB2008_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 52) landscape raster."

"LAND" "PCTSHRB2008WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2008.tif (class 52) landscape raster."

"LAND" "PCTSHRB2011_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 52) landscape raster."

"LAND" "PCTSHRB2011WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2011.tif (class 52) landscape raster."

"LAND" "PCTSHRB2013_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 52) landscape raster."

"LAND" "PCTSHRB2013WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2013.tif (class 52) landscape raster."

"LAND" "PCTSHRB2016_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 52) landscape raster."

"LAND" "PCTSHRB2016WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2016.tif (class 52) landscape raster."

"LAND" "PCTSHRB2019_WSRP100" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 52) landscape raster."

"LAND" "PCTSHRB2019WS" "% of the upstream watershed (Ws) areas classified as shrub/scrub land cover. Derived from the nlcd2019.tif (class 52) landscape raster."

"LAND" "PCTSILICICWS" "% of the upstream watershed (Ws) classified as lithology type: silicic residual material."

"LAND" "PCTUNCONSOL_SAND_GRAVELWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Unconsolidated sand and gravel."

"LAND" "PCTURBHI2001_WSRP100" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 24) landscape raster."

"LAND" "PCTURBHI2001WS" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use. Derived from the nlcd2001.tif (class 24) landscape raster."

"LAND" "PCTURBHI2004_WSRP100" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 24) landscape raster."

"LAND" "PCTURBHI2004WS" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use. Derived from the nlcd2004.tif (class 24) landscape raster."

"LAND" "PCTURBHI2006_WSRP100" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 24) landscape raster."

"LAND" "PCTURBHI2006WS" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use. Derived from the nlcd2006.tif (class 24) landscape raster."

"LAND" "PCTURBHI2008_WSRP100" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 24) landscape raster."

"LAND" "PCTURBHI2008WS" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use. Derived from the nlcd2008.tif (class 24) landscape raster."

"LAND" "PCTURBHI2011_WSRP100" "% of the upstream watershed (Ws) areas classified as developed, high intensity land use. Derived from the nlcd2011.tif (class 24) landscape raster."

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space land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 21) landscape raster."

"LAND" "PCTURBOP2016WS" "% of the upstream watershed (Ws) areas classified as developed, open space land use. Derived from the nlcd2016.tif (class 21) landscape raster."

"LAND" "PCTURBOP2019_WSRP100" "% of the upstream watershed (Ws) areas classified as developed, open space land use and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 21) landscape raster."

"LAND" "PCTURBOP2019WS" "% of the upstream watershed (Ws) areas classified as developed, open space land use. Derived from the nlcd2019.tif (class 21) landscape raster."

"LAND" "PCTVOLCANICWS" "% of the upstream watershed (Ws) classified as bedrock permeability type: Basalt and other volcanic rocks."

"LAND" "PCTWATERWS" "% of the upstream watershed (Ws) classified as lithology type: water."

"LAND" "PCTWDWET2001_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2001.tif (class 90) landscape raster."

"LAND" "PCTWDWET2001WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2001.tif (class 90) landscape raster."

"LAND" "PCTWDWET2004_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2004.tif (class 90) landscape raster."

"LAND" "PCTWDWET2004WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2004.tif (class 90) landscape raster."

"LAND" "PCTWDWET2006_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2006.tif (class 90) landscape raster."

"LAND" "PCTWDWET2006WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2006.tif (class 90) landscape raster."

"LAND" "PCTWDWET2008_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2008.tif (class 90) landscape raster."

"LAND" "PCTWDWET2008WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2008.tif (class 90) landscape raster."

"LAND" "PCTWDWET2011_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2011.tif (class 90) landscape raster."

"LAND" "PCTWDWET2011WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2011.tif (class 90) landscape raster."

"LAND" "PCTWDWET2013_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2013.tif (class 90) landscape raster."

"LAND" "PCTWDWET2013WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2013.tif (class 90) landscape raster."

"LAND" "PCTWDWET2016_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2016.tif (class 90) landscape raster."

"LAND" "PCTWDWET2016WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2016.tif (class 90) landscape raster."

"LAND" "PCTWDWET2019_WSRP100" "% of the upstream watershed (Ws) areas classified as woody wetland land cover and within 100-m buffer of NHD stream lines (Rp100). Derived from the nlcd2019.tif (class 90) landscape raster."

"LAND" "PCTWDWET2019WS" "% of the upstream watershed (Ws) areas classified as woody wetland land cover. Derived from the nlcd2019.tif (class 90) landscape raster."

"LAND" "PERDUN_WS_PCTFULL" "Percent of area with data for Dunne overland flow, also know as saturation overland flow, is generated in a basin when the water table \outcrops\\" on the land surface (due to the infiltration and redistribution of soil moisture within the basin), thereby producing temporary saturated areas. These saturated areas generate Dunne overland flow through exfiltration of shallow ground water and by routing precipitation directly to the stream network. The data are estimated from a watershed simulation model called TOPMODEL (Wolock, 1993). The model was run throughout the conterminous U.S. and estimated percentage of Dunne overland flow in total streamflow was quantified (Wolock, 2003).\""

"LAND" "PERDUNWS" "Dunne overland flow, also know as saturation overland flow, is generated in a basin when the water table \outcrops\\" on the land surface (due to the infiltration and redistribution of soil moisture within the basin), thereby producing temporary saturated areas. These saturated areas generate Dunne overland flow through exfiltration of shallow ground water and by routing precipitation directly to the stream network. The data are estimated from a watershed

simulation model called TOPMODEL (Wolock, 1993). The model was run throughout the conterminous U.S. and estimated percentage of Dunne overland flow in total streamflow was quantified (Wolock, 2003).\"

"LAND" "PERHOR_WS_PCTFULL" "Percent of area with data for Horton overland flow, also known as infiltration-excess overland flow, is generated in a basin when infiltration rates are exceeded by precipitation rates. The data are estimated from a watershed simulation model called TOPMODEL (Wolock, 1993). The model was run throughout the conterminous U.S. and estimated percentage of Horton overland flow in total streamflow was quantified (Wolock, 2003)."

"LAND" "PERHORWS" "Horton overland flow, also known as infiltration-excess overland flow, is generated in a basin when infiltration rates are exceeded by precipitation rates. The data are estimated from a watershed simulation model called TOPMODEL (Wolock, 1993). The model was run throughout the conterminous U.S. and estimated percentage of Horton overland flow in total streamflow was quantified (Wolock, 2003)."

"LAND" "PERM_WS_PCTFULL" "Percent of area with data for Mean of all perm values within upstream watershed (Ws)."

"LAND" "PERMH_WS_PCTFULL" "Percent of area with data for Watershed mean of the high values of permeability (inches/hour) of soils from the State Soil Geographic (STATSGO) Database"

"LAND" "PERMHWS" "Watershed mean of the high values of permeability (inches/hour) of soils from the State Soil Geographic (STATSGO) Database"

"LAND" "PERMWS" "Mean of all perm values within upstream watershed (Ws)."

"LAND" "PESTIC97_WS_PCTFULL" "Percent of area with data for Mean of all pestic.tif values within upstream watershed (Ws)."

"LAND" "PESTIC97WS" "Mean of all pestic.tif values within upstream watershed (Ws)."

"LAND" "PMAX_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $(MAXxi) / 30$, where xi = the modeled total precipitation (mm) for month i (1-12). Values based on 30 years (1971-2000) of PRISM climate estimates. Each value represents a 900 x 900 meter cell. Mean of all Pmax values within the watershed upstream of the sampling point ($?Pmax\ i / n$, where i = each of n pixels within the watershed)."

"LAND" "PMAWS" "GIS raster calculated as $(MAXxi) / 30$, where xi = the modeled total precipitation (mm) for month i (1-12). Values based on 30 years (1971-2000) of PRISM climate estimates. Each value represents a 900 x 900 meter cell. Mean of all Pmax values within the watershed upstream of the sampling point ($?Pmax\ i / n$, where i = each of n pixels within the watershed)."

"LAND" "PMIN_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $(MINxi) / 30$, where xi = the modeled total precipitation (mm) for month i (1-12). Values based on 30 years (1971-2000) of PRISM climate estimates. Each value represents a 900 x 900 meter cell. Mean of all Pmin values within the watershed upstream of the sampling point ($?Pmin\ i / n$, where i = each of n pixels within the watershed)."

"LAND" "PMINWS" "GIS raster calculated as $(MINxi) / 30$, where xi = the modeled total precipitation (mm) for month i (1-12). Values based on 30 years (1971-2000) of PRISM climate estimates. Each value represents a 900 x 900 meter cell. Mean of all Pmin values within the watershed upstream of the sampling point ($?Pmin\ i / n$, where i = each of n pixels within the watershed)."

"LAND" "POPDEN2010_WS_PCTFULL" "Percent of area with data for Mean of all popden2010 values within the upstream watershed (Ws)."

"LAND" "POPDEN2010_WS_PCTFULLRP100" "Percent of area with data for Mean of all popden2010 values within the upstream watershed (Ws) within a 100-m wide buffer of the NHD stream lines (Rp100). Derived as the intersection between Rp100 and popden2010 landscape layers."

"LAND" "POPDEN2010WS" "Mean of all popden2010 values within the upstream watershed (Ws)."

"LAND" "POPDEN2010WSRP100" "Mean of all popden2010 values within the upstream watershed (Ws) within a 100-m wide buffer of the NHD stream lines (Rp100). Derived as the intersection between Rp100 and popden2010 landscape layers."

"LAND" "PRECIP_WS_PCTFULL" "Percent of area with data for mean annual precip (cm) for the watershed, from 800m PRISM data. 30 years period of record 1971-2000. "

"LAND" "PRECIP08WS" "PRISM climate data - mean precipitation (mm):, within the upstream watershed (Ws). This metric was derived to support modeling of stream biological condition based on the 2008-2009 NRSA."

"LAND" "PRECIP09WS" "PRISM climate data - mean precipitation (mm):, within the upstream watershed (Ws). This metric was derived to support modeling of stream biological condition based on the 2008-2009 NRSA."

"LAND" "PRECIP8110WS" "PRISM climate data - 30-year average annual normal precipitation (mm):, within the upstream watershed (Ws) "

"LAND" "PRISMPPT_2008_WS_PCTFULL" "Percent of area with data for PRISM climate data - mean precipitation (mm):, within the upstream watershed (Ws). This metric was derived to support modeling of stream biological condition based on the 2008-2009 NRSA."

"LAND" "PRISMPPT_2009_WS_PCTFULL" "Percent of area with data for PRISM climate data - mean precipitation (mm):, within the upstream watershed (Ws). This metric was derived to support modeling of stream biological condition based on the 2008-2009 NRSA."

[illegible]

represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LANDSCAPE" "PSUMPY_2014_PT" "GIS raster value at site location calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2014_WS_PCTFULL" "Percent of area with data for GIS raster calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2014WS" "GIS raster calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2018_PT" "GIS raster value at site location calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2018_WS_PCTFULL" "Percent of area with data for GIS raster calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2018WS" "GIS raster calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2019_PT" "GIS raster value at site location calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2019_WS_PCTFULL" "Percent of area with data for GIS raster calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "PSUMPY_2019WS" "GIS raster calculated as x_i , where x_i = the modeled total precipitation (mm) for month i (5-12) of the previous year and month i (1-4) of the sampling year. This value estimates the total precipitation for the 12 months prior to the field sampling season. Each value represents a 2 x 2 kilometer cell (<http://www.prism.oregonstate.edu>)."

"LAND" "QTSDEVOL" "Percentage of the watershed covered by any of the geology types QTRNRY, SDMNTRY, or VOLCANIC. Values 5, 6, or 7. (Reed & Bush, 2001)"

"LAND" "RCKDEP_WS_PCTFULL" "Percent of area with data for Mean of all rckdep values within the upstream watershed (Ws)."

"LAND" "RCKDEPWS" "Mean of all rckdep values within the upstream watershed (Ws)."

"LAND" "RDCRS_WS_PCTFULL" "Percent of area with data for Sum of all rdstrcrs values within the upstream watershed (Ws) divided by the area of each Cat and Ws."

"LAND" "RDCRSSLPWTDWS" "Sum of all rdstrcrs values within the upstream watershed (Ws) multiplied by NHDPlusV21 slope and divided by the area of each Cat and Ws."

"LAND" "RDCRSWS" "Sum of all rdstrcrs values within the upstream watershed (Ws) divided by the area of each Cat and Ws."

"LAND" "RDDENS_WS_PCTFULL" "Percent of area with data for Mean of all rddens values within the upstream watershed (Ws)."

"LAND" "RDDENS_WS_PCTFULLRP100" "Percent of area with data for Density of roads (2010 Census Tiger Lines) within watershed and within a 100-m buffer of NHD stream lines (km/square km)"

"LAND" "RDDENWS" "Mean of all rddens values within the upstream watershed (Ws)."

"LAND" "RDDENWSRP100" "Density of roads (2010 Census Tiger Lines) within watershed and within a 100-m buffer of NHD stream lines (km/square km)"

"LANDSCAPE" "RDH_PT" "Site mean of the high values of depth to bedrock (inches) of soils from the State Soil Geographic (STATSGO) Database."

"LAND" "RDH_WS_PCTFULL" "Percent of area with data for Site mean of the high values of depth to bedrock (inches) of soils from the State Soil Geographic (STATSGO) Database."

"LAND" "RDHWS" "Site mean of the high values of depth to bedrock (inches) of soils from the State Soil Geographic (STATSGO) Database."

"LANDSCAPE" "RHMEAN_PT" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled mean relative humidity (%) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell "

"LAND" "RHMEAN_WS_PCTFULL" "Percent of area with data for Watershed average relative humidity (percent), from 2km PRISM, derived from 30 years of record (1961-1990)."

"LAND" "RHMEANWS" "Watershed average relative humidity (percent), from 2km PRISM, derived from 30 years of record (1961-1990)."

"LAND" "RPU" "Raster Processing Unit that the site falls in, from NHDPlusV21."

"LAND" "RUNOFF_WS_PCTFULL" "Percent of area with data for Mean of all runoff values within the upstream watershed (Ws)."

"LAND" "RUNOFFWS" "Mean of all runoff values within the upstream watershed (Ws)."

"LAND" "S_WS_PCTFULL" "Percent of area with data for Mean of all S values within the upstream watershed (Ws)."

"LAND" "SAND_WS_PCTFULL" "Percent of area with data for Mean of all sand values within the upstream watershed (Ws)."

"LAND" "SANDWS" "Mean of all sand values within the upstream watershed (Ws)."

"LAND" "SIO2_WS_PCTFULL" "Percent of area with data for Mean of all SiO2 values within the upstream watershed (Ws)."

"LAND" "SIO2WS" "Mean of all SiO2 values within the upstream watershed (Ws)."

"VERIFY" "SITE_ID" "Site ID"

"DESIGN" "SITE_ID" "Site ID for the study and year sampled. Preferred form is NRSnn_MM-xxxxx where nn is current cycle year (04, 08, 13, 18, etc), MM is two letter state code, and xxxxx is between 10001 and 99999. No missing or \"blank\" values allowed."

"LANDSCAPE" "SITE_ID" "Unique Site ID assigned for NRSA13-14"

"LAND" "SN_2008_WS_PCTFULL" "Percent of area with data for Annual average of precipitation-weighted mean deposition for average sulfur & nitrogen concentration for year, 2008, within the upstream watershed (Ws)."

"LAND" "SN_2008WS" "Annual average of precipitation-weighted mean deposition for average sulfur & nitrogen concentration for year, 2008, within the upstream watershed (Ws)."

"LAND" "SUPERFUND_WS_PCTFULL" "Percent of area with data for density of georeferenced Superfund sites (Superfund.shp) within the upstream watershed (Ws)."

"LAND" "SUPERFUND_WS_PCTFULLRP100" "Percent of area with data for density of georeferenced Superfund sites (Superfund.shp) within the upstream watershed (Ws) within a 100-m buffer (Rp100) of the NHD stream lines. Derived as the intersection between RpRp100 and Superfund.shp landscape layers."

"LAND" "SUPERFUNDDENSWS" "Density of georeferenced Superfund sites (Superfund.shp) within the upstream watershed (Ws)."

"LAND" "SUPERFUNDDENSWSRP100" "Density of georeferenced Superfund sites (Superfund.shp) within the upstream watershed (Ws) within a 100-m buffer (Rp100) of the NHD stream lines. Derived as the intersection between RpRp100 and Superfund.shp landscape layers."

"LAND" "SWS" "Mean of all S values within the upstream watershed (Ws)."

"LAND" "TMAX_WS_PCTFULL" "Percent of area with data for PRISM climate data - 30-year average annual normal maximum air temperature (C°);, within the upstream watershed (Ws) "

"LAND" "TMAX8110WS" "PRISM climate data - 30-year average annual normal maximum air temperature (C°);, within the upstream watershed (Ws) "

"LAND" "TMEAN_PW_2004_PT" "GIS raster value at site location calculated as $x_i / 3$, where x_i = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (x_i) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2004_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $x_i / 3$, where x_i = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (x_i) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2004WS" "GIS raster calculated as $x_i / 3$, where x_i = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (x_i) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2005_PT" "GIS raster value at site location calculated as $x_i / 3$, where x_i = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the

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temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2014_PT" "GIS raster value at site location calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2014_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2014WS" "GIS raster calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2018_PT" "GIS raster value at site location calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2018_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2018WS" "GIS raster calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2019_PT" "GIS raster value at site location calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2019_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_PW_2019WS" "GIS raster calculated as $xi / 3$, where xi = the modeled mean air temperature (°C) for month i (12) of the previous year and month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the winter just prior to when field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_S_2004_PT" "GIS raster value at site location calculated as $xi / 4$, where xi = the modeled mean air temperature (°C) for month i (6-9) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell "

"LAND" "TMEAN_S_2004_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi /$

"LAND" "TMEAN_S_2013_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $\sum_{i=1}^4 x_i / 4$, where x_i = the modeled mean air temperature ($^{\circ}\text{C}$) for month i (6-9) of the sample year. The

"LAND" "TMEAN_SY_2004_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $\frac{x_i}{12}$, where x_i = the modeled mean air temperature ($^{\circ}\text{C}$) for month i (1-12) of the sample year. The modeled monthly mean air temperature (x_i) is the average of the minimum and maximum monthly air

"LAND" "TMEAN_SY_2013_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $\frac{x_i}{12}$, where x_i = the modeled mean air temperature ($^{\circ}\text{C}$) for month i (1-12) of the sample year. The modeled monthly mean air temperature (x_i) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average

[illegible]

[illegible]

"LAND" "TMEAN_W_2013WS" "GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2014_PT" "GIS raster value at site location calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2014_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2014WS" "GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2018_PT" "GIS raster value at site location calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2018_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2018WS" "GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2019_PT" "GIS raster value at site location calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2019_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_W_2019WS" "GIS raster calculated as $xi / 4$, where xi = the modeled mean air temperature ($^{\circ}C$) for month i (1-2) of the sample year. The modeled monthly mean air temperature (xi) is the average of the minimum and maximum monthly air temperatures (<http://www.prism.oregonstate.edu/faq.phtml>). This value estimates the average temperature of the specific summer that field sampling was done. Each value represents a 2 x 2 kilometer cell"

"LAND" "TMEAN_WS_PCTFULL" "Percent of area with data for PRISM climate data - 30-year average annual normal mean air temperature ($^{\circ}C$):, within the upstream watershed (Ws) "

"LAND" "TMEAN8110WS" "PRISM climate data - 30-year average annual normal mean air temperature ($^{\circ}C$):, within the upstream watershed (Ws) "

"LAND" "TMIN_WS_PCTFULL" "Percent of area with data for PRISM climate data - 30-year average annual normal minimum air temperature ($^{\circ}C$):, within the upstream watershed (Ws) "

"LAND" "TMIN8110WS" "PRISM climate data - 30-year average annual normal minimum air temperature ($^{\circ}C$):, within the upstream watershed (Ws) "

"LAND" "TRI_WS_PCTFULL" "Percent of area with data for density of georeferenced Toxic Release Inventory sites (TRI.shp) within the upstream watershed (Ws)."

"LAND" "TRI_WS_PCTFULLRP100" "Percent of area with data for density of georeferenced Toxic Release Inventory sites (TRI.shp) within the upstream watershed (Ws) within a 100-m buffer (Rp100) of the NHD

stream lines. Derived as the intersection between RpRp100 and TRI.shp landscape layers."

"LAND" "TRIDENSWWS" "Density of georeferenced Toxic Release Inventory sites (TRI.shp) within the upstream watershed (Ws)."

"LAND" "TRIDENSWSRP100" "Density of georeferenced Toxic Release Inventory sites (TRI.shp) within the upstream watershed (Ws) within a 100-m buffer (Rp100) of the NHD stream lines. Derived as the intersection between RpRp100 and TRI.shp landscape layers."

"UNIQUE_ID" "Unique site ID based on location that does not depend on year sampled or any other factor. Form is NRS_ss_nnnnn where ss state and nnnnn is number that starts at 10001. No missing values allowed."

"LAND" "VPU" "Vector Processing Unit that the site falls in, from NHDPlusV21."

"LANDSCAPE" "WDMAX_PT" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled maximum number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell\""

"LAND" "WDMAX_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled maximum number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell\""

"LAND" "WDMAXWS" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled maximum number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell\""

"LANDSCAPE" "WDMIN_PT" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled minimum number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell\""

"LAND" "WDMIN_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled minimum number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell\""

"LAND" "WDMINWS" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled minimum number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell\""

"LANDSCAPE" "WDSUM_PT" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled mean number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell \""

"LAND" "WDSUM_WS_PCTFULL" "Percent of area with data for GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled mean number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell \""

"LAND" "WDSUMWS" "GIS raster calculated as $(\text{SUM}x_i / 12) / 30$, where x_i = the modeled mean number of days with measurable precipitation (i.e., \wetdays\\", days) for month i (1-12). Values based on 30 years (1961-1990) of PRISM climate estimates. Each value represents a 2 x 2 kilometer cell \""

"LAND" "WETINDEXWS" "Mean Composite Topographic Index (CTI)[Wetness Index]:, within the upstream watershed (Ws) "

"LAND" "WETNESSINDEX_WS_PCTFULL" "Percent of area with data for Mean Composite Topographic Index (CTI)[Wetness Index]:, within the upstream watershed (Ws) "

"LANDSCAPE" "WSAREASQKM" "Area in square kilometers of the watershed basin. Summarized by joining the sub-catchment area of the catchment with all of the upstream catchments of the NHDPlusV21."

"LAND" "WSAREASQKMRP100" "Area in square kilometers within 100-m buffer of NHD stream lines (Rp100) of the watershed basin. Summarized by joining the sub-catchment riparian zone area of the catchment with all of the upstream catchment's riparian zone areas."

"LAND" "WTDEP_WS_PCTFULL" "Percent of area with data for Mean of all wtdep values within the upstream watershed (Ws)."

"LAND" "WTDEPWS" "Mean of all wtdep values within the upstream watershed (Ws)."